

# Chapter 3.1

An Exponential function is in the form  $f(x) = a^x$  where  $a$  is a constant Note:  $f(x) = x^a$  is not exponential  
 called the base power function  
 important

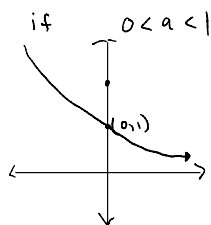
Let  $x = n$  a positive integer

then  $a^x = a^n = \underbrace{a \cdot a \cdot \dots \cdot a}_{n \text{ times}}$

if  $x = 0$  then  $a^0 = 1$

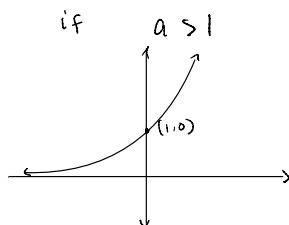
recall  $\sqrt[n]{a^p} = a^{\frac{p}{n}}$  where  $p, n$  are integers :  $n > 0$  :  $a^x = \frac{1}{a^{-x}}$

## Graphs



$$\lim_{x \rightarrow \infty} a^x = 0$$

$$\lim_{x \rightarrow -\infty} a^x = \infty$$



$$\lim_{x \rightarrow \infty} a^x = \infty$$

$$\lim_{x \rightarrow -\infty} a^x = 0$$

Note exponential function with base  $a$ ,  $a > 1$  grows faster than any polynomial

Ex  $\lim_{x \rightarrow \infty} (4^x + 3) = \infty + 3 = \infty$

$\rightarrow 2^{-x} = \frac{1}{2^x}$   $\lim_{x \rightarrow \infty} \frac{1}{2^x} = \frac{1}{\infty} = 0$  is  $2^\infty$  a big or small #

Ex  $\lim_{x \rightarrow \infty} \frac{2^{-x} + 2}{5 + 4^{-x}} = \frac{\lim_{x \rightarrow \infty} 2^{-x} + 2}{5 + \lim_{x \rightarrow \infty} 4^{-x}}$   
 $\rightarrow 4^{-x} = \frac{1}{4^x}$   $\lim_{x \rightarrow \infty} \frac{1}{4^x} = 0$   
 $= \frac{2}{5}$

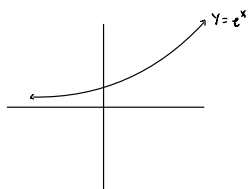
Ex  $\lim_{x \rightarrow \infty} \frac{6+3^x}{2-5^x} = \frac{\infty}{\infty}$  find another way

$$\lim_{x \rightarrow \infty} \frac{6+3^x}{2-5^x} \cdot \frac{\frac{1}{3^x}}{\frac{1}{3^x}} = \lim_{x \rightarrow \infty} \frac{\frac{6}{3^x} + 1}{\frac{2}{5^x} - 1} = -1$$

"e"

The # "e" is defined as  $e \approx 2.71828 \dots$ , irrational

$f(x) = e^x$



recall  $\lim_{x \rightarrow \infty} e^{-6x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = \frac{1}{\infty} = 0$

Ex  $\lim_{x \rightarrow \infty} \frac{(e^{3x} + 2e^{-3x}) \cdot \frac{1}{e^x}}{2e^{-x} + 3e^{3x} \cdot \frac{1}{e^x}} = \lim_{x \rightarrow \infty} \frac{1 + 2e^{-6x}}{2e^{-4x} + 3} = \frac{1}{3}$

Side Note

$$\frac{e^{3x}}{e^{3x}} = e^{3x-3x} = e^0 = 1$$

$$\frac{e^{-3x}}{e^{3x}} = e^{-3x-3x} = e^{-6x}$$

$$\frac{e^x}{e^{3x}} = e^{-x-3x} = e^{-4x}$$